of general relativity, he would have predicted the expansion of the universe more than a decade before it was discovered observationally. That would certainly have ranked among the greatest discoveries—it might have been *the* greatest discovery—of all time. After learning of Hubble's results, Einstein rued the day he had thought of the cosmological constant, and he carefully erased it from the equations of general relativity. He wanted everyone to forget the whole sorry episode, and for many decades everyone did.

In the 1980s, however, the cosmological constant resurfaced in a surprising new form and ushered in one of the most dramatic upheavals in cosmological thinking since our species first engaged in cosmological thought.

Of Jumping Frogs and Supercooling

If you caught sight of a baseball flying upward, you could use Newton's law of gravity (or Einstein's more refined equations) to figure out its subsequent trajectory. And if you carried out the required calculations, you'd have a solid understanding of the ball's motion. But there would still be an unanswered question: Who or what threw the ball upward in the first place? How did the ball acquire the initial upward motion whose subsequent unfolding you've evaluated mathematically? In this example, a little further investigation is all it generally takes to find the answer (unless, of course, the aspiring big-leaguers realize that the ball just hit is on a collision course with the windshield of a parked Mercedes). But a more difficult version of a similar question dogs general relativity's explanation of the expansion of the universe.

The equations of general relativity, as originally shown by Einstein, the Dutch physicist Willem de Sitter, and, subsequently, Friedmann and Lemaître, allow for an expanding universe. But, just as Newton's equations tell us nothing about how a ball's upward journey got started, Einstein's equations tell us nothing about how the expansion of the universe got started. For many years, cosmologists took the initial outward expansion of space as an unexplained given, and simply worked the equations forward from there. This is what I meant earlier when I said that the big bang is silent on the bang.